

Jason Appiah-Kubi

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EDUCATION

University of Maryland, Baltimore County

Catonsville, MD

Bachelor of Science in Computer Science

May 2025

- **Relevant Coursework:** Data Structures, Cloud Computing, Operating Systems, Software Engineering

EXPERIENCE

Contract Website Developer

September 2024 – January 2025

The Hidden Parts Podcast

Catonsville, MD

- Developed and maintained a responsive website using HTML, CSS, React, and node.js
- Led client requirement gathering sessions, resulting in 100% on-time delivery and exceeding quality benchmarks

Software Engineer Intern

May 2024 - August 2024

Sonavex Inc.

Baltimore, MD

- Architected and implemented a new technology stack that reduced data processing time by 30% through optimized algorithms and efficient database design
- Developed a medical imaging analysis tool using Python and PyQt, implementing a memory-efficient data model with NumPy arrays and spatial indexing for fast scan retrieval and processing.
- Collaborated with medical professionals to design and implement responsive GUI features based on user feedback
- Refactored a legacy MATLAB codebase to Python, saving over \$4,000 annually in licensing fees.

Software Engineer Intern

May 2023 - August 2023

US Devcom Army Research Lab

Aberdeen Proving Ground, MD

- Developed solutions for multi-agent robotic simulations by integrating Unity and Unreal Engine through an SQL-based data bridge.
- Designed and implemented a Graphical User Interface in RVIZ, enhancing the visualization and control of robotic simulations.
- Utilized a diverse technology stack (C#, Unity, Unreal, ROS 1/2, C++, Python, QT, RVIZ, SQL) to improve simulation fidelity and robotic testing capabilities.

PROJECTS

Task Sync | *JavaScript, Node.js, Express.js, React, MongoDB, WebSockets*

August 2024 – Present

- Developed a full-stack web application built with JavaScript. It serves a REST API using Express.js on the backend and uses React for the frontend.
- Designed and implemented a MongoDB database schema to manage user data, tasks, and group information, ensuring optimal data retrieval and scalability
- Implemented real-time updates for group collaboration using WebSockets, ensuring all users see live changes to tasks and schedules.

Developer Documentation Search Engine | *Next.js, TypeScript, PostgreSQL, Redis*

October 2024 - Present

- Engineered a full-stack web based search engine with Next.js & Node.js, implementing Elasticsearch-powered indexing, fuzzy search, and real-time query optimization.
- Designed a PostgreSQL database to store user data, bookmarks, and search history, with Redis caching for optimized performance.
- Developed documentation indexing with support for multiple sources (MDN, React, TypeScript), using web scraping and API integrations.

Music Genre Classifier | *Python, Pytorch, numpy, matplotlib, Kaggle, Librosa*

January 2025 - Present

- Developed a machine learning model to classify music genres based on audio features extracted from sound files.
- Trained the model on a large dataset containing hundreds of labeled audio samples, enhancing classification performance and generalization.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, Kotlin, Ruby

Frameworks: React, Node.js, TypeScript, JUnit, Tailwind, Bootstrap, ROS 1/2, Express.js, Qt, PyQt

Developer Tools: Git, Docker, Figma, RVIZ, PostgreSQL, MySQL, MongoDB, AWS, Linux, JUnit, Linux, Jira

Libraries: Pandas, NumPy, Matplotlib, Librosa, Tailwind CSS